

APP β -CTF triggers cell-autonomous synaptic toxicity independent of A β

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eLife Assessment

This study presents a **useful** demonstration that a specific protein fragment may induce the loss of synapses in Alzheimer's disease. The evidence supporting the data is **solid** but only partially supports the conclusion and would benefit from additional discussion indicated by the literature from reviewer #1. The application of the findings is limited because blocking the formation of the protein fragment has not benefited patients in several clinical trials.

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Abstract

A β is believed to play a significant role in synaptic degeneration observed in Alzheimer's disease (AD) and is primarily investigated as a secreted peptide. However, the contribution of intracellular A β or other cleavage products of its precursor protein (APP) to synaptic loss remains uncertain. In this study, we conducted a systematic examination of their cell-autonomous impact using a sparse expression system. Here, these proteins/peptides were overexpressed in a single neuron, surrounded by thousands of untransfected neurons. Surprisingly, we found that APP induced dendritic spine loss only when co-expressed with BACE1. This effect was mediated by β -CTF, a β -cleavage product of APP, through an endosome-related pathway independent of A β . Neuronal expression of β -CTF in mouse brains resulted in defective synaptic transmission and cognitive impairments, even in the absence of amyloid plaques. These findings unveil a β -CTF-initiated mechanism driving synaptic toxicity irrespective of amyloid plaque formation and suggest a potential intervention by inhibiting the endosomal GTPase Rab5.

Introduction

Alzheimer's disease (AD) stands as the most prevalent form of neurodegenerative disease and a leading cause of dementia. It is characterized by extracellular amyloid plaque deposition, intracellular neurofibrillary tangles, synaptic and neuronal loss, and neuroinflammation (Blennow et al., 2006 ; Gómez-Isla et al., 1996 ; John Hardy and Selkoe, 2002 ; Leyns and Holtzman,

2017 [\[1\]](#); Selkoe, 2002 [\[2\]](#)). The amyloid plaque primarily consists of aggregated A β , a cleavage product of amyloid precursor protein (APP) (Glenner and Wong, 1984 [\[3\]](#); Masters et al., 1985 [\[4\]](#)). Notably, two A β antibodies have demonstrated efficacy in removing amyloid plaques and slowing AD progression in phase III clinical trials (Sims et al., 2023 [\[5\]](#); van Dyck et al., 2023 [\[6\]](#)). Furthermore, numerous naturally occurring mutations in genes encoding APP or its catalytic enzyme γ -secretase result in early onset familial AD or reduce AD risk (Jonsson et al., 2012 [\[7\]](#); Liu et al., 2017 [\[8\]](#); Mullan et al., 1992 [\[9\]](#); Nilsberth et al., 2001 [\[10\]](#)). This collective evidence, spanning AD pathology, human genetics, and intervention trials, strongly supports a causal role of A β and amyloid plaque in AD pathogenesis. However, despite clinical trials employing A β antibodies targeting A β oligomers, protofibrils, or deposited plaque, AD progression has been slowed down by only ~30% (Mintun et al., 2021 [\[11\]](#); Pleen and Townley, 2022 [\[12\]](#); Sims et al., 2023 [\[5\]](#); van Dyck et al., 2023 [\[6\]](#)). Notably, antibodies against monomeric soluble A β failed to yield clinical benefits (Sperling et al., 2023 [\[13\]](#)). It is possible that these A β antibodies may overlook certain pathogenic factors crucial for AD pathogenesis.

APP, a type I transmembrane protein, undergoes cleavage primarily by α -secretase on the cytoplasmic membrane, producing soluble α -cleavage N-terminal fragment (sAPP α) and α -cleavage C-terminal fragment (α -CTF). Some APP molecules bypass α -cleavage and undergo endocytosis into endocytic compartments, where they are subsequently cleaved by β -secretase, generating soluble β -cleavage N-terminal fragment (sAPP β) and β -cleavage C-terminal fragment (β -CTF). β -CTF is further cleaved by γ -secretase to produce A β and APP intracellular domain (AICD) (Golde et al., 1992 [\[14\]](#); Zhang and Song, 2013 [\[15\]](#); Zhang et al., 2011 [\[16\]](#)). While extracellular A β , targeted by A β antibodies, is widely studied, the potential contribution of intracellular A β , APP, and other APP cleavage products to AD pathogenesis remains uncertain (Konietzko, 2012 [\[17\]](#); Kwart et al., 2019 [\[18\]](#); Nikolaev et al., 2009 [\[19\]](#); Oddo et al., 2003 [\[20\]](#); Vohra et al., 2010 [\[21\]](#); Willem et al., 2015 [\[22\]](#)). For instance, β -CTF has been implicated in endosomal dysfunction (Israel et al., 2012 [\[23\]](#); Jiang et al., 2010 [\[24\]](#); Kim et al., 2016 [\[25\]](#); Kwart et al., 2019 [\[18\]](#); Xu et al., 2016 [\[26\]](#)), yet its downstream functional impacts remain unclear.

Synapse loss represents an early feature of AD neurodegeneration and is closely associated with cognitive dysfunction (de Wilde et al., 2016 [\[27\]](#); DeKosky et al., 1996 [\[28\]](#); Terry et al., 1991 [\[29\]](#)). Secreted A β induces synaptic dysfunction by interacting with its receptors on the neuronal plasma membrane (Kamenetz et al., 2003 [\[30\]](#); Kessels et al., 2013 [\[31\]](#); Wei et al., 2010 [\[32\]](#)). Other studies have reported γ -secretase inhibition reduced spine density *in vivo* via an APP-dependent pathway (Bittner et al., 2009 [\[33\]](#)). Additionally, deletion of APP in mice has been shown to decrease dendritic spine density (Tyan et al., 2012 [\[34\]](#)). The diverse outcomes of APP on synapse suggest a complex impact of APP and metabolites. It remains unclear whether APP or other APP fragments can also induce synaptic toxicity in a cell-autonomous manner.

To address these inquiries, we employed a sparse transfection system utilizing the Helios gene gun to explore the potential role of intracellular A β or other APP fragments. These molecules were expressed in a single neuron surrounded by untransfected wild type neurons. Surprisingly, full-length APP did not induce synaptic toxicity. However, co-expression of APP with BACE1 resulted in significant loss of dendritic spines, indicating a crucial role of APP β -cleavage in synaptic damage. Further investigations unveiled that this detrimental effect was mediated by β -CTF in a cell-autonomous manner, independent of A β . Additionally, *in vivo* expression of β -CTF was adequate to induce synaptic dysfunction and cognitive impairments in mice, even in the absence of amyloid plaques. In summary, our study delineates a mechanism initiated by β -CTF that can induce synaptic degeneration in a cell-autonomous manner, thus extending beyond the scope of A β antibody-based therapies.

Results

APP only led to spine loss when co-expressed with BACE1

To investigate the cell-autonomous impact of APP on neurons, we utilized a Helios gene gun transfection system to sparsely express APP in rat organotypic hippocampal slice cultures (Chen et al., 2014). In hippocampal slice cultures at DIV9, transient expression of a familial AD APP mutation (APP_{Swedish}) (Mullan et al., 1992) with GFP in CA1 pyramidal neurons for six days did not reduce their dendritic spine densities compared with neurons expressing GFP alone (Figure 1A-B), suggesting that APP alone does not significantly contribute to synaptic loss.

BACE1-mediated β -cleavage of APP is crucial for amyloidogenesis, and BACE1 activity is notably elevated in AD patient brains (Cheng et al., 2014; Vassar et al., 1999). In human brains, APP and BACE1 are expressed at a ratio of about 15:1 (Uhlén et al., 2015). Endogenous BACE1 levels may not suffice to cleave the overexpressed APP. To assess whether insufficient β -cleavage of APP underlies the lack of synaptic toxicity caused by overexpressed APP, we co-expressed APP with BACE1 (in a 15:1 ratio) in organotypic hippocampal slice cultures and observed a significant ~40% reduction in spine density compared to neurons expressing GFP alone (Figure 1C-D). Transient expression of BACE1 alone did not affect spine density (Figure 1E-F). Co-expression of APP and BACE1 with different ratios or using an internal ribosome entry site (IRES) also resulted in significant spine loss (Supplementary figure 1A-B). These findings support the requirement of BACE1 for APP to induce synaptic loss.

When expressed alone in HEK293T cells, APP is predominantly cleaved at its α site to produce sAPP α and α -CTF (C-terminal fragment after APP α -cleavage) (Figure 1G, Supplementary figure 1C). Co-expression with BACE1 led to increased cleavage of APP at its β and β' sites, resulting in elevated β -CTF (more than 4-fold) and β' -CTF as the major metabolic products and a substantial reduction in α -CTF levels (Figure 1G-H, Supplementary figure 1C). APP_{MV} (M596V) cannot be cleaved by BACE1 to produce β -CTF and A β but has no impact on β' -cleavage (Figure 1I-K) (Citron et al., 1995). When co-expressed with BACE1, APP_{MV} failed to induce spine loss (Figure 1L-M), supporting the requirement of β -cleavage of APP to induce spine loss.

In summary, these findings suggest that certain β -cleavage products of APP, rather than APP or BACE1 alone, could lead to spine loss in a cell-autonomous manner.

β -CTF induced spine loss independent of A β

APP undergoes cleavage by secretases, generating soluble N-terminal fragments (sAPPs) and C-terminal fragment (CTFs). We then explored the impact of different sAPPs and APP-CTFs on dendritic spines. Transient expression of β -CTF significantly reduced the spine density of CA1 pyramidal neurons in organotypic rat hippocampal cultures, whereas expression of sAPP α , sAPP β , α -CTF or β' -CTF did not produce such an effect (Figure 2A-B, Supplementary figure 2A-B). APP CTFs exhibited similar expression levels in HEK293T cells (Figure 2C). When the plasmid amount was reduced to 1/8 of the original dose, β -CTF no longer induced a decrease in dendritic spine density (Supplementary figure 2E-F), indicating the synaptic damaging effect of β -CTF is its expression level dependent.

β -CTF undergoes processing by γ -secretase to generate A β and AICD (APP intracellular domain) (Thinakaran and Koo, 2008). Subsequently, we investigated whether A β generated from β -CTF was responsible for the β -CTF-induced spine loss. Treatment with a γ -secretase inhibitor, PF03084014 (1 μ M), effectively reduced A β to baseline levels in cells expressing β -CTF without altering the expression levels of β -CTF itself (Figure 2D-E). PF03084014 treatment did not affect the spine density in neurons expressing GFP alone (Supplementary figure 2C-D). Notably,

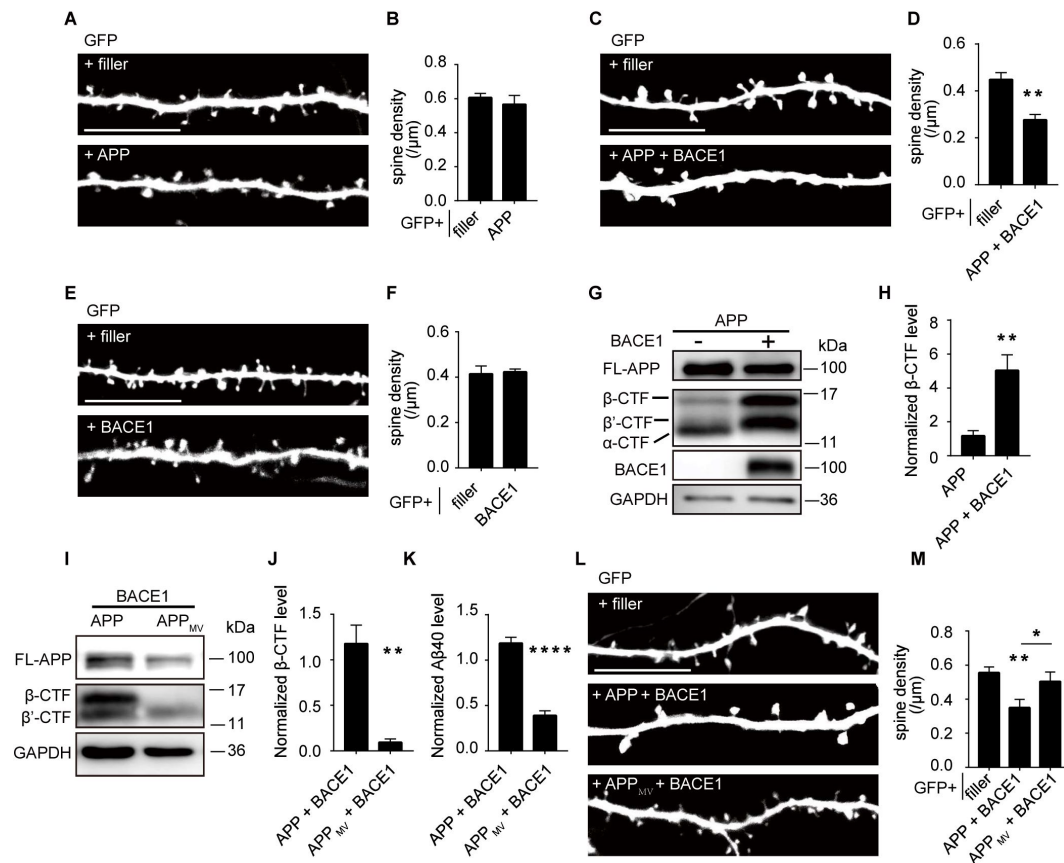


Figure 1.

APP only led to spine loss when co-expressed with BACE1.

A-B Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP alone or together with APP. (filler, $n=21$; APP, $n=16$).

C-D Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP alone or together with APP and BACE1 with a ratio of 15:1. (filler, $n=8$; APP plus BACE1, $n=5$)

E-F Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP alone or together with BACE1. (filler, $n=15$; BACE1, $n=15$)

G Western blot and corresponding statistical analysis of APP (Y188), BACE1 and GAPDH from HEK293T cells expressing APP or APP plus BACE1 with a ratio of 15:1. GAPDH was measured as a loading control.

H Measurement of APP β -CTF (β C-Terminal Fragment) level from HEK293T cells expressing APP alone or APP and BACE1 with a ratio of 15:1. $n=4$.

I Western blot and corresponding statistical analysis of APP fragments (Y188) from HEK293T cells expressing APP plus BACE1 or APP_{MV} plus BACE1.

J Measurement of APP β -CTF levels from HEK293T cells expressing APP plus BACE1 or APP_{MV} plus BACE1. $n=4$.

K Measurement of A β 40 levels from HEK293T cells expressing APP plus BACE1 or APP_{MV} plus BACE1. $n=4$.

L-M Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP alone or together with APP and BACE1 or APP_{MV} and BACE1. (filler, $n=12$; APP plus BACE1, $n=14$; APP_{MV} plus BACE1, $n=10$)

All dendritic images were acquired from rat organotypic hippocampal slice cultures after transfection for 6-7 days. Statistics: One-way ANOVA or Student's t test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Error bars show SEM. Scale bars, 10 μ m.

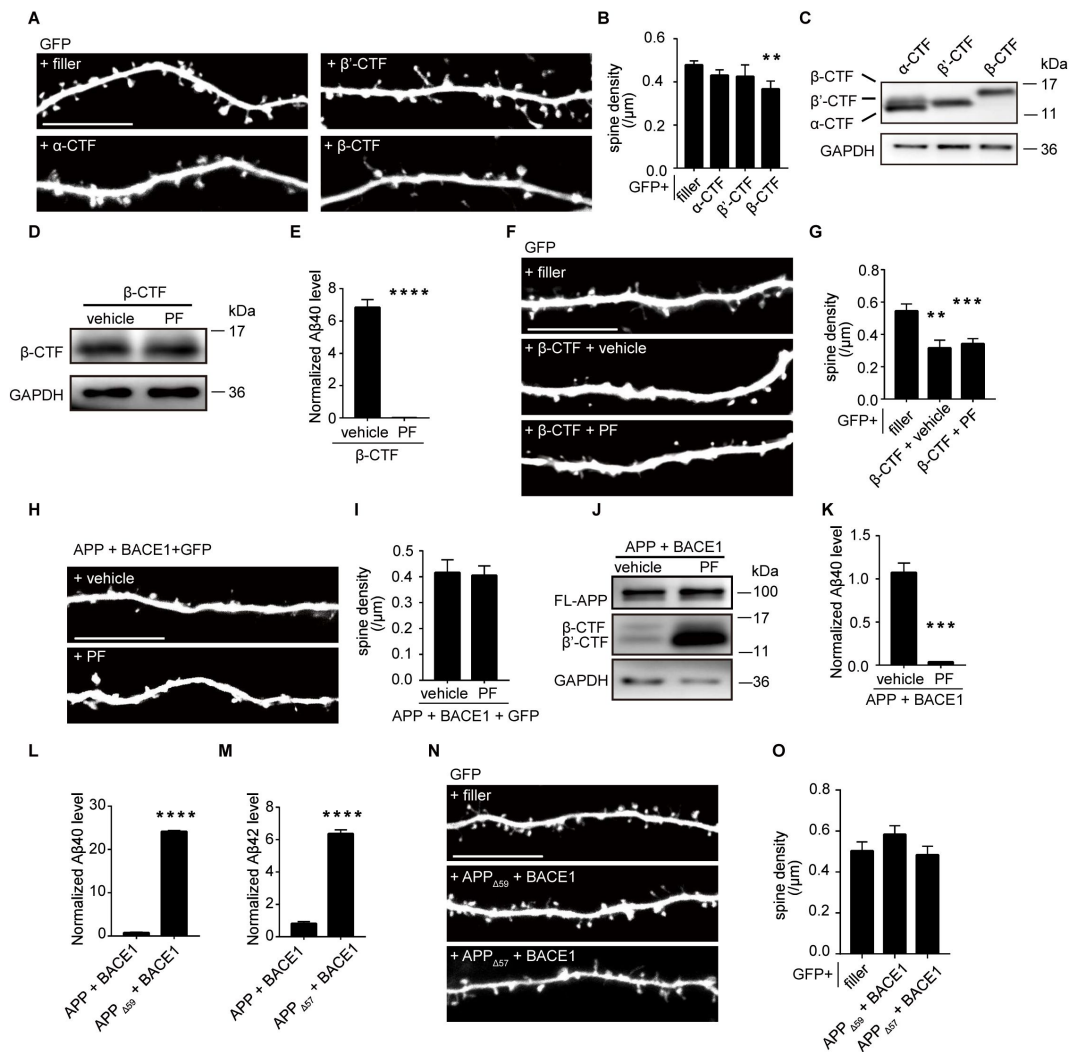


Figure 2.

β-CTF induced spine loss independent of Aβ.

A-B Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP alone or together with α-CTF, β'-CTF or β-CTF. (filler, n=19; α-CTF, n=7; β'-CTF, n=8; β-CTF, n=7)

C Western blot of APP fragments from HEK293T cells expressing α-CTF-flag, β'-CTF-flag and β-CTF-flag (flag antibody).

D Western blot of APP fragments from HEK293T cells expressing β-CTF-flag treatment with vehicle or PF (flag antibody).

E Measurement of Aβ40 secreted from HEK293T cells expressing β-CTF treatment with vehicle or PF. n=4.

F-G Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP alone or together with β-CTF after treated with vehicle or PF. (filler, n=11; β-CTF, n=7; β-CTF plus PF, n=14)

H-I Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP together with APP and BACE1 after treated with vehicle or PF. (APP plus BACE1 with vehicle, n=9; APP plus BACE1 with PF, n=14)

J Western blot of APP fragments (Y188) from HEK293T cells co-expressing APP and BACE1 after treated with vehicle or PF.

K Measurements of Aβ40 from HEK 293T cells expressing APP and BACE1 after treated with vehicle or PF. n=4.

L Measurements of Aβ40 from HEK293T cells expressing APP and BACE1 or APP_{Δ59} and BACE1. n=4.

M Measurements of Aβ42 from HEK293T cells expressing APP and BACE1 or APP_{Δ57} and BACE1. n=4.

N-O Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP alone or together with APP_{Δ59/Δ57} and BACE1. (filler, n=9; APP_{Δ59} and BACE1, n=7; APP_{Δ57} and BACE1, n=6)

All dendritic images were acquired from rat organotypic hippocampal slice cultures after transfection for 6-7 days. PF, PF03084014, a γ secretase inhibitor. Statistics: One-way ANOVA or Student's t test. * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001. Error bars show SEM. Scale bars, 10 μm.

PF03084014 treatment failed to prevent the spine loss induced by β -CTF expression (**Figure 2F-G**), or by co-expression of APP and BACE1 (**Figure 2H-K**), suggesting that A β might not be the causative factor.

To delve deeper into the impact of A β on dendritic spines, we engineered two APP mutants (APP $_{\Delta 59}$ and APP $_{\Delta 57}$) lacking the AICD domain, which are known to generate significant amounts of A β 40 and A β 42, respectively (**Figure 2L-M**). Co-expression of either APP $_{\Delta 59}$ or APP $_{\Delta 57}$ with BACE1, did not alter spine density (**Figure 2N-O**), further bolstering the idea that A β is not responsible for the β -CTF-induced spine loss. Subsequently, we investigated the involvement of another β -CTF cleavage product, AICD, which has been reported to interact with the transcription factor forkhead box O (FoxO) and promote FoxO-induced transcription of proapoptotic genes, leading to cell death (Wang et al., 2014). However, transient expression of AICD failed to alter the density of spines (Supplementary figure 2G-H).

In conclusion, these results support the idea that APP can induce spine loss in a cell-autonomous manner through β -CTF, independent of A β and AICD.

Expression of β -CTF damaged synapses in mice

To ascertain whether β -CTF-induced spine loss could manifest *in vivo*, we examined the density of dendritic spines from CA1 pyramidal neurons infected with lentivirus encoding β -CTF and GFP, or GFP alone, in adult mice. Sparse expression of GFP and β -CTF was observed in CA1 pyramidal neurons (**Figure 3A**). The spine density of CA1 pyramidal neurons expressing β -CTF and GFP was significantly lower than that of neurons expressing GFP alone (**Figure 3B-C**), indicating that β -CTF could induce spine loss in a cell-autonomous manner *in vivo*.

Next, we evaluated excitatory synaptic transmission through whole-cell patch-clamp recording of hippocampal pyramidal neurons infected with lentivirus encoding β -CTF or GFP. Neurons expressing β -CTF exhibited a ~65% lower frequency of mEPSCs compared to neighboring uninfected neurons, whereas there was no significant change in mEPSC frequency in neurons expressing GFP alone (**Figure 3D-E**). However, mEPSC amplitude remained unaltered in neurons expressing either β -CTF or GFP (**Figure 3D, F**). These findings collectively suggest that β -CTF leads to reduced excitatory synapse density without affecting the strength of the remaining synapses.

Expression of β -CTF damaged cognitive function in mice in the absence of plaque formation

We proceeded to investigate whether β -CTF affects cognitive functions. Adeno-associated viruses (AAV) encoding GFP or β -CTF were bilaterally injected into the hippocampus of 1-month-old mice (**Figure 4A-B**). After 4 months of expression, animal behaviors were examined. Immunostaining revealed widespread expression of GFP or β -CTF throughout the entire hippocampus (**Figure 4B**), with no detectable amyloid plaque formation observed using ThS staining (**Figure 4C**). Side-by-side staining demonstrated robust amyloid plaque deposition in the brain of two-month-old 5XFAD mice (**Figure 4D**), validating the efficacy of the staining method. The body weight of mice expressing β -CTF in the hippocampus was approximately 17% lower than that of GFP controls (**Figure 4E**). In the Y-maze test, the mean alternations were significantly reduced in mice expressing β -CTF compared to GFP controls, suggesting abnormal working memory in mice expressing β -CTF (**Figure 4F**).

The fear conditioning test assesses associative fear learning and memory (**Figure 4G**) (Xiao et al., 2018). In this test, mice expressing β -CTF exhibited similar baseline levels of freezing time as GFP controls (**Figure 4H**). However, in the hippocampus-dependent contextual fear conditioning test, mice expressing β -CTF showed a significantly shorter freezing time (approximately 40% less) than those expressing GFP after training (**Figure 4H**). In the amygdala-

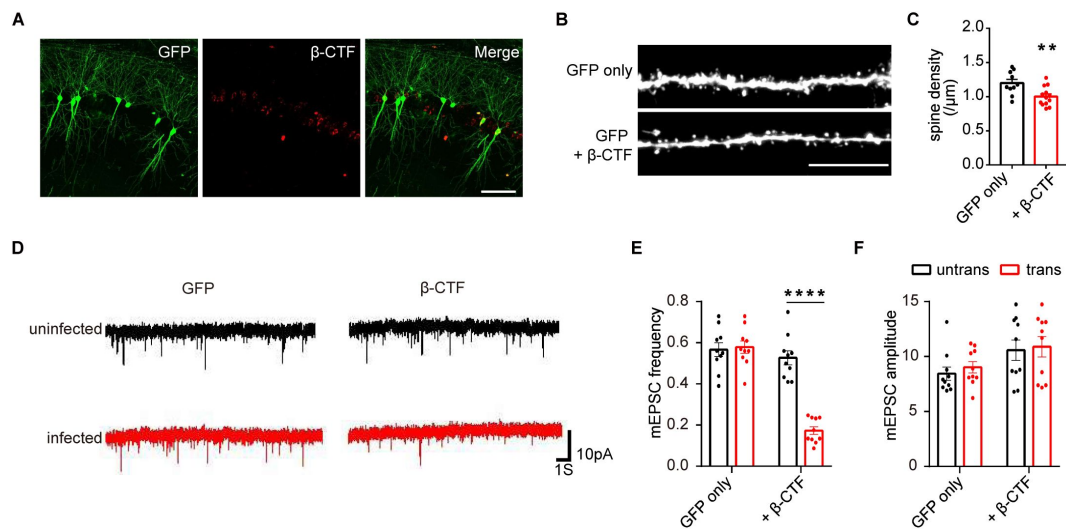


Figure 3.

Expression of β -CTF damaged synapses in mouse brains.

A Representative images of adult mouse CA1 pyramidal neurons infected with lentivirus expressing GFP and β -CTF. Scale bar, 100 μ m.

B-C Representative images from basal dendrites and quantitation of their spine density in CA1 pyramidal neurons infected with lentivirus expressing GFP alone or GFP with β -CTF in mouse hippocampi. Scale bar, 10 μ m. GFP only, n=10; GFP plus β -CTF, n=13.

D-F Representative recording traces and quantitations of mEPSCs from adult mouse hippocampal neurons infected with lentivirus (identified with GFP signal) and neighboring uninfected neurons. n=10.

Statistics: Two-way ANOVA or Student's t test. * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001. Error bars show SEM.

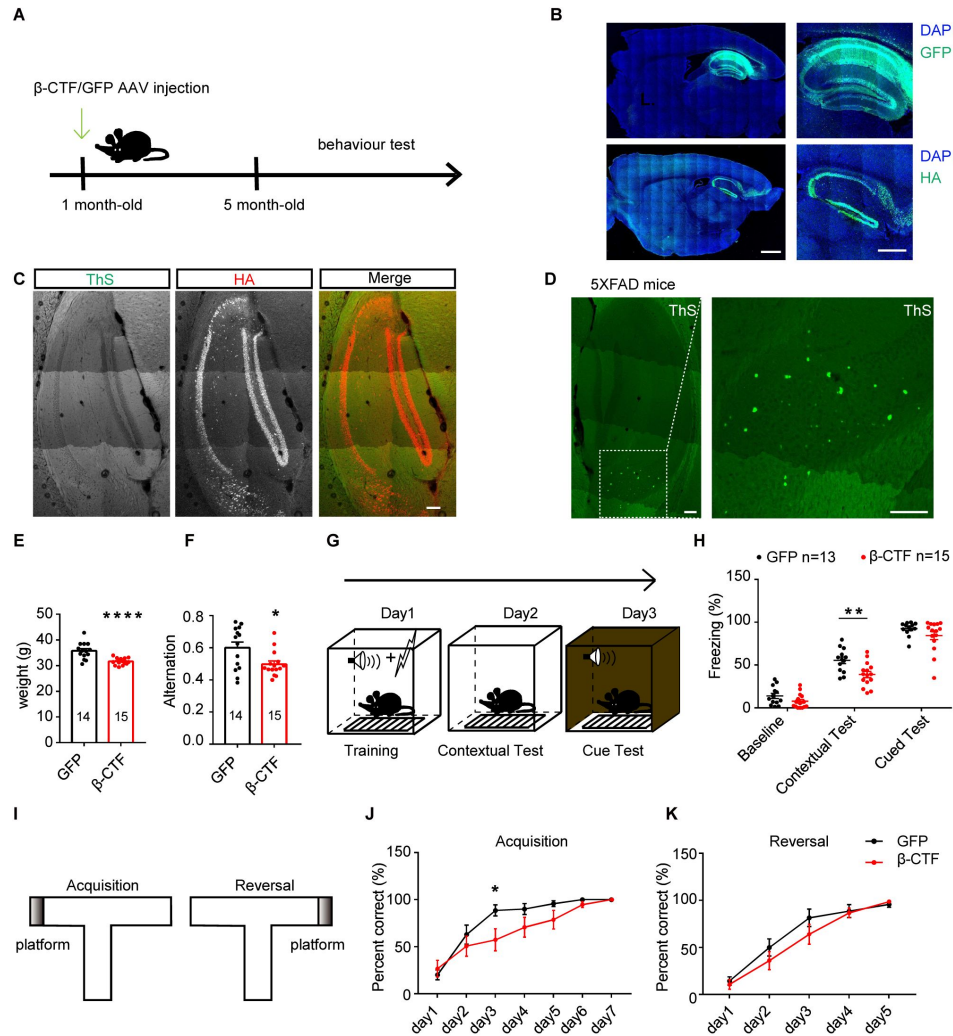


Figure 4.

Expression of β -CTF damaged cognitive function in mice in the absence of plaque formation.

A Schematic diagram showing the time line for stereotactic injection of AAV, behavioral training and tests.

B Representative images showing immunofluorescence staining of expressed GFP or β -CTF-HA by AAV in mouse hippocampi (WT). Scale bars represent 1000 μ m (left) and 500 μ m (right).

C Images of mouse hippocampus (WT) stained with ThS and HA antibody after infected with AAV encoding β -CTF-HA. Scale bar, 100 μ m.

D Representative image of ThS staining of a 5XFAD mice hippocampal slice. Scale bar, 100 μ m.

E Weight of mice infected with AAV encoding GFP or β -CTF in their hippocampi. GFP, n=14; β -CTF, n=15.

F Quantitations of spontaneous alternation in Y-maze from mice infected with AAV encoding GFP or β -CTF. GFP, n=14; β -CTF, n=15.

G Schematic diagram showing fear conditioning test design.

H Quantitation of freezing in contextual and cued fear conditioning tests from mice infected with AAV encoding GFP and β -CTF in their hippocampi. GFP, n=13; β -CTF, n=15.

I Schematic diagram showing water T maze test design.

J-K The percentage of correct responses across the 5 trials of water T maze test was quantified in each day of Acquisition (J) or Reversal (K). GFP, n=14; β -CTF, n=15.

Statistics: Repeated measures two-way ANOVA or Student's t test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Error bars show SEM.

dependent cued fear conditioning test, these two groups performed similarly (**Figure 4H**). In the water T maze test (**Figure 4I**), mice expressing β -CTF displayed slower learning curves than GFP controls, indicating impairments in acquisition learning (**Figure 4J**) but not in reversal learning (**Figure 4K**).

In the open field test, mice expressing β -CTF traveled a longer distance (Supplementary figure 3A-C) and spent more time in the center area with a higher frequency of center entrances (Supplementary figure 3D-E) compared to mice expressing GFP. However, there were no differences in rearing frequency and duration between the two groups (Supplementary figure 3F-G). These results suggest that mice expressing β -CTF exhibited increased locomotion and reduced anxiety-like behaviors. Consistently, mice injected with AAV encoding β -CTF spent significantly more time exploring and traveled a greater distance in the open arms compared to the GFP controls in the EPM (Supplementary figure 3H-J). In tail suspension tests, these two groups exhibited similar levels of immobility (Supplementary figure 3K), indicating that β -CTF expression in the hippocampus did not alter depression-like behaviors.

In conclusion, these findings support that β -CTF expression is sufficient to disrupt hippocampus-dependent cognitive functions.

The C-terminal YENPTY motif was necessary for β -CTF to induce endosomal dysfunction and synapse loss

Endosome abnormalities mediated by APP β -CTFs have been reported across various cell types, including human iPSC-induced neurons, PC12M cells, and N2a cells (Kim et al., 2016; Kwart et al., 2019; Xu et al., 2016). Next, we investigated whether β -CTF impacted endosomes in hippocampal neurons. Dissociated cultured hippocampal neurons were co-transfected with APP-CTFs and Rab5-GFP, an endosomal marker fused with GFP, at DIV7. In neurons expressing either Rab5-GFP alone or Rab5-GFP with α -CTF, Rab5 puncta appeared uniform and smoothly rounded (**Figure 5A-B**). However, in neurons co-transfected with β -CTF, Rab5 puncta were larger and exhibited less uniform shapes, often appearing lobular, and showed robust co-localization with β -CTF (**Figure 5C**, D). Notably, the morphology of lysosomes, as observed by Lamp1 staining, remained unaffected by the expression of α - or β -CTF in neurons (Supplementary figure 4A-D), thereby suggesting a specific interaction of β -CTF with endosomes.

The C-terminal YENPTY motif of APP was found to be crucial for its interaction with endosomes (Lai et al., 1995). To elucidate the significance of this interaction, we expressed a mutant form of β -CTF (β -CTF_{mut}), where the YENPTY motif was substituted with AENATA. Interestingly, β -CTF_{mut} was expressed at similar levels to wildtype β -CTF and exhibited comparable A β production (**Figure 5E-H**). However, unlike wildtype β -CTF, β -CTF_{mut} showed a more diffuse distribution throughout the neurons and failed to induce enlarged Rab5 puncta (**Figure 5I-J**). Notably, pyramidal neurons transiently expressing β -CTF_{mut} displayed a higher spine density compared to those expressing wildtype β -CTF (**Figure 5K-L**). These findings underscore the critical role of the YENPTY motif-mediated interaction with endosomes in β -CTF-induced spine loss.

Spine loss induced by β -CTF was prevented by Rab5 inhibition

To investigate the downstream mechanism responsible for β -CTF-induced synaptic loss, we analyzed the proteomic alterations triggered by β -CTF in dissociated cultured hippocampal neurons (**Figure 6A**). Our findings revealed significant changes in protein levels upon exposure to β -CTF (**Figure 6B**). Notably, the expression of Synapsin-1, a presynaptic protein associated with synaptic vesicles, and GluR1, GluN2A, GluN2B, subunits of glutamate receptors, were diminished in neurons expressing β -CTF (**Figure 6C-F**). VAMP2, a key component of SNARE complex was also reduced by β -CTF (**Figure 6E**, G). The protein level of Synapsin-1 was further reduced after treatment with γ -secretase inhibitors in neurons expressing β -CTF (Supplementary figure 5A-B). Gene Ontology analysis further elucidated that β -CTF expression downregulated

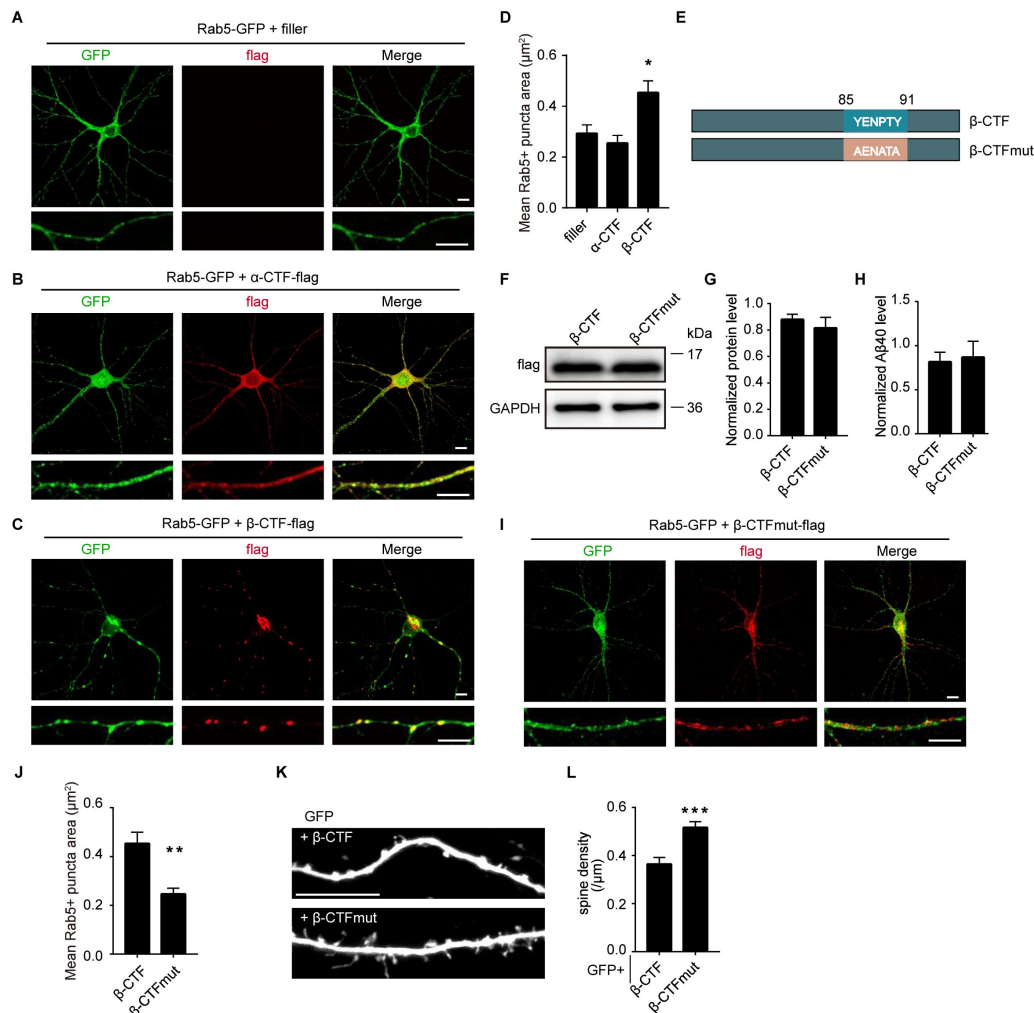


Figure 5.

The C-terminal YENPTY motif was necessary for β-CTF to induce endosomal dysfunction and synapse loss.

A Immunofluorescent staining of GFP (green) from dissociated rat hippocampal neurons expressing Rab5-GFP.

B Immunofluorescent staining of GFP (green) and flag (red) from dissociated rat hippocampal neurons expressing Rab5-GFP and α-CTF-flag.

C Immunofluorescent staining of GFP (green) and flag (red) from dissociated rat hippocampal neurons expressing Rab5-GFP and β-CTF-flag.

D Quantitation of average Rab5+ puncta size of neurons expressing Rab5-GFP only or co-expressing α/β-CTF-flag and Rab5-GFP. (filler, n=9; α-CTF, n=9; β-CTF, n=11)

E Schematic diagram of the β-CTF and β-CTFmut.

F Western blot of APP fragments from HEK293T cells expressing β-CTF-flag and β-CTFmut-flag with a flag antibody.

G Quantitations of APP fragments from HEK293T cells expressing β-CTF-flag and β-CTFmut-flag. n=4.

H Measurement of Aβ40 from HEK 293T cells expressing β-CTF or β-CTFmut. n=5.

I Immunofluorescent staining of GFP (green) and flag (red) from dissociated rat hippocampal neurons expressing Rab5-GFP and β-CTFmut-flag.

J Quantitation of average Rab5+ puncta size of neurons co-expressing β-CTF-flag or β-CTFmut-flag and Rab5-GFP. (β-CTF, n=11; β-CTFmut, n=9)

K-L Representative images and spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP together with β-CTF or β-CTFmut for 6-7 days in rat organotypic hippocampal slice cultures. β-CTF, n=12; β-CTFmut, n=15. Scale bar, 10 μm. Statistics: One-way ANOVA or Student's t test. * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001. Error bars show SEM.

proteins involved in membrane trafficking, synaptic vesicle cycle, pre-synapse, and post-synapse (**Figure 6H-I**), aligning with previous observations indicating that β -CTF induces synaptic dysfunction and endosomal abnormalities (Zhou et al., 2019). Conversely, upregulated proteins were primarily associated with peptide metabolic processes and translation (Supplementary figure 5C-D).

Overexpression of β -CTF resulted in the enlargement of Rab5-positive endosomes, similar to the effects observed with a constitutively active mutant of Rab5, Rab5^{Q79L} (Supplementary figure 5E-F) (Kim et al., 2016). Notably, the introduction of a dominant negative mutant of Rab5, Rab5^{S34N}, attenuated the endosome enlargement induced by β -CTF (**Figure 6J-K**). In neurons expressing Rab5^{S34N}, there was reduced co-localization of β -CTF with Rab5 (**Figure 6J-K**). Critically, co-expression of Rab5^{S34N} with β -CTF effectively mitigated the spine loss induced by β -CTF in hippocampal slice cultures (**Figure 6L-M**). These findings underscored that Rab5 overactivation-induced endosomal dysfunction contributed to β -CTF-induced spine loss. However, expression of Rab5^{S34N} in β -CTF-expressing neurons did not alter the levels of synapse-related proteins that were reduced in these neurons (Supplementary figure 5G-H), suggesting Rab5 overactivation did not contribute to these protein expression changes induced by β -CTF.

Discussion

Treatment with Lecanemab or Donanemab, two A β antibodies, significantly slowed down AD progression by approximately 30% in phase III clinical trials (Sims et al., 2023; van Dyck et al., 2023). The success of these antibodies in modifying AD progression validates A β as a cause of AD pathogenesis. However, the relatively modest benefits have raised additional questions. Despite effectively reducing amyloid plaques to near baseline levels after 18 months of treatment in some patients, functional benefits were limited to only ~30%. This raises the question of whether amyloid plaque is the primary source of A β -related toxicity or if these patients were treated too late to reverse their disease progression more effectively. This study aims to investigate whether A β -associated pathways could lead to pathogenesis beyond secreted A β and amyloid plaques, which falls outside the scope of A β antibody-based therapies due to their inability to penetrate cytoplasmic membranes.

AD is characterized by significant synaptic degeneration. Conventionally, A β is considered detrimental to synapses, thus synaptic loss in AD is attributed to various forms of A β , including amyloid plaques (Kamenetz et al., 2003; Oakley et al., 2006). However, conflicting studies have suggested that A β may actually promote synaptogenesis (Abramov et al., 2009; Bittner et al., 2009; Puzzo et al., 2008; Zhou et al., 2022). The exact role of A β in AD-related synaptic degeneration remains elusive. Most investigations have focused on the non-cell-autonomous function of A β after its secretion from neurons (Reinders et al., 2016). The potential contribution of intracellular A β or its precursors to synaptic toxicity has been largely unexplored. Utilizing a sparse neuron transfection system, we systematically examined this question. Intriguingly, among A β , APP, and major APP cleavage products, only β -CTF induced synaptic toxicity in a cell-autonomous manner (**Figure 1**, 2, Supplementary figure 1, 2), aligning with predictions based on indirect evidence (Kwart et al., 2019; Lee et al., 2022; Tamayev et al., 2012; Vaillant-Beuchot et al., 2021; Xu et al., 2016). In mouse brains, β -CTF also triggered substantial synaptic loss and cognitive deficits in the absence of amyloid plaques (**Figure 4C**), further supporting the notion that synaptic loss and amyloid plaque formation are mediated by distinct mechanisms. While current data did not exclude the potential involvement of A β -induced toxicity in the synaptic and cognitive dysfunction observed in mice overexpressing β -CTF, addressing this directly remains challenging. Treatment with γ -secretase inhibitors could potentially shed light on this issue. However, treatments with γ -secretase inhibitors are known to lead to brain dysfunction

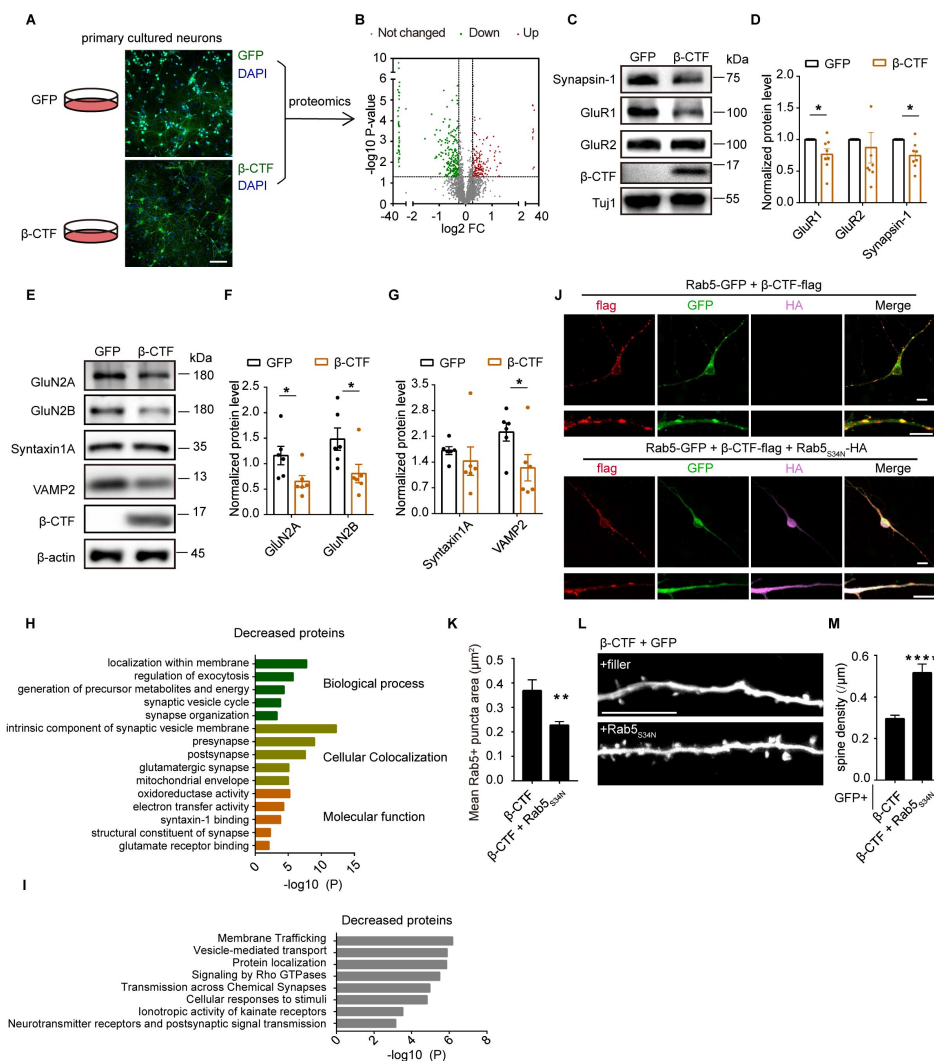


Figure 6.

Spine loss induced by β -CTF was prevented by Rab5 inhibition.

A Representative images of dissociated rat hippocampal neurons infected with lenti virus encoding GFP or β -CTF. Scale bar, 100 μ m.

B Volcano plot of quantitative mass spectrometry analysis showing protein omics changes in neurons infected with lenti virus encoding β -CTF v.s. GFP. Proteins with fold change >1.2 or <0.83 and p value <0.05 are colored.

C-D Western blot and corresponding statistical analysis of Synapsin-1, GluR1, GluR2 in dissociated hippocampal neurons infected with lentivirus expressing GFP or β -CTF. Tuj 1 was used as an internal control.

E-G Western blot and corresponding statistical analysis of GluN2A, GluN2B, Syntaxin1A, VAMP2 in dissociated hippocampal neurons infected with lentivirus expressing GFP or β -CTF. β -actin was used as an internal control.

H The Gene Ontology (GO) processes analysis of decreased proteins by Metascape.

I The Reactome Gene Sets processes analysis of decreased protein by Metascape.

J Immunofluorescent staining of GFP (green), flag (red) and HA (magenta) from dissociated rat hippocampal neurons expressing Rab5-GFP and β -CTF-flag or together with Rab5_{S34N}-HA. Scale bar, 10 μ m.

K Quantitation of average Rab5+ puncta size from dissociated rat hippocampal neurons expressing Rab5-GFP and β -CTF-flag or together with Rab5_{S34N}-HA. (β -CTF, n=5; β -CTF + Rab5_{S34N}, n=9)

L-M Representative images and measurements of spine density of basal dendrites from CA1 pyramidal neurons transiently expressing GFP and β -CTF with or without Rab5_{S34N} for 6-7 days in rat organotypic hippocampal slice cultures. Scale bar, 10 μ m. (β -CTF, n=17; β -CTF plus Rab5_{S34N}, n=11)

Statistics: Student's t test. * p <0.05 , ** p <0.01 , *** p <0.001 , **** p <0.0001 . Error bars show SEM.

by itself likely due to its blockade of the γ -cleavage of other essential molecules, such as Notch (Doody et al., 2013 [DOI](#); Güner and Lichtenthaler, 2020 [DOI](#)), preventing from pursuing it further experimentally *in vivo*.

APP, A β , and presenilins have been extensively studied in mouse models, providing convincing evidence that high A β concentrations are toxic to synapses (Chapman et al., 1999 [DOI](#)). Moreover, addition of A β to murine cultured neurons or brain slices is toxic to synapses (Wang et al., 2017 [DOI](#)). However, A β -induced synaptotoxicity was not observed in our study. A major difference between our study and others is that we employed an isolated expression system, where A β was applied solely to individual neurons surrounded by other neurons, without overwhelming them with excessive amounts of A β . In contrast, other studies typically apply A β to neurons indiscriminately. Therefore, we predict that A β does not lead to synaptic deficits from individual neurons in cell autonomous manners, whereas β -CTF does.

It is noteworthy that β -CTF induces spine loss independent of A β , implying that while synaptic degeneration and amyloidogenesis both occur downstream of β -CTF, they may represent two parallel processes independent of each other following APP β -cleavage. In fact, A β and β -CTF levels were significantly elevated in the AD brains (Kim et al., 2016 [DOI](#)). A β antibodies effectively facilitate the clearance of secreted A β , including that deposited in amyloid plaques. However, β -CTF localizes within the intracellular membrane of endo/lysosomal vesicles, rendering it inaccessible to A β antibodies. Consequently, A β antibody-based therapies cannot mitigate the neuronal toxicities initiated by β -CTF. This limitation could partially explain the restricted clinical benefits observed with A β antibodies. Addressing β -CTF-mediated synaptic toxicity should be prioritized in the future development of improved AD therapies, potentially in conjunction with A β antibodies.

However, clinical trials of BACE1 inhibitors have failed to demonstrate an efficacy in treating AD, despite their theoretical potential to inhibit the production of both β -CTF and A β . One potential reason is that pharmacological inhibition of BACE1 appears not as effective as its genetic removal. Genetic depletion of BACE1 leads to clearance of existing amyloid plaques (Hu et al., 2018 [DOI](#)), whereas pharmacological inhibition of BACE1 could not stop growth of existing plaques, although it slows down the growth of these plaques and prevents formation of new plaques (Peters et al., 2018 [DOI](#)). The development of better β -secretase inhibitors to more effectively inhibit β -cleavage is required to fully test the role of APP β -cleavage in AD pathogenesis.

Endosomal dysfunction is an early neuropathological signature of AD and Down syndrome (DS) mediated by β -CTF. Kim et al found β -CTF recruited APPL1 (a Rab5 effector) via YENPTY motif to Rab5 endosomes, where it stabilizes active GTP-Rab5, leading to pathologically accelerated endocytosis, endosome swelling and selectively impaired transport of Rab5 endosomes (Kim et al., 2016 [DOI](#)). Rab5 endosomal dysfunction may trigger prodromal and neurodegenerative features of Alzheimer's disease (Pensalfini et al., 2020 [DOI](#)). However, the downstream functional impacts of endosomal dysfunction induced by β -CTF remain unclear. One of the impacts is deficits of NGF signaling in basal forebrain cholinergic neurons (BFCNs) (Salehi et al., 2006 [DOI](#); Xu et al., 2016 [DOI](#)). Here, we report endosomal dysfunction-induced by β -CTF also contributes to synaptic dysfunction. Synaptic abnormalities are corrected by inhibiting Rab5 activity to restore endosomal function (**Figure 6J-M** [DOI](#)), indicating that endosomal abnormalities may serve as a potential target for AD treatment.

β -CTF elicited significant spine loss, whereas α -CTF or β' -CTF, which are only 16 or 10 amino acids shorter at their N-terminal ends compared to β -CTF, did not exhibit this detrimental effect. Since the C-terminal YENPTY motif of β -CTF is essential for this activity, likely through its interaction with endosomal proteins, β -CTF represents the minimum size of the APP fragment capable of inducing cell-autonomous synaptic toxicity. However, the specific contribution of the N-terminal domain of β -CTF to synaptic loss remains unclear. It remains to be investigated whether longer

fragments containing β -CTF could also induce synaptic loss. If confirmed, strategies aimed at reducing APP β -cleavage while promoting α -or β' -cleavage to decrease both A β and β -CTF may offer a more effective treatment for AD in the future.

Materials and methods

Dissociated hippocampal neuron culture and transfection

We prepared hippocampal neurons from embryonic day 17/18 (E17/18) SD rats. Initially, whole hippocampi were dissected under stereomicroscopes and then digested into single cells using papain (Worthington) for 25 minutes at 37 °C. The dissociated cells were seeded at a density of 180,000 cells per well in a 24-well plate, using Neurobasal media (Thermo Fisher Scientific) supplemented with 2% B27 (Life technology), 0.25% Glutamax (Thermo Fisher Scientific), and 100 units/mL Penicillin/Streptomycin (Life technology). Neurons were cultured in incubators maintained at 37 °C with 5% CO₂, with fresh media half replenished every 5 days. For virus infection, dissociated hippocampal neurons at day *in vitro* (DIV) 14 were exposed to purified lentivirus. After 5 days, the infected neurons were lysed in RIPA buffer (Solarbio) supplemented with a protease inhibitor cocktail (Selleck) for further analysis. For plasmid transfection, dissociated hippocampal neurons at DIV7 were transfected with Lipofectamine 2000 (Thermo Fisher Scientific) following the manufacturer's instructions.

Western blotting and immunofluorescent staining

For western blotting, we first removed the culture medium, and then immediately placed neurons or HEK293T cells on ice. Subsequently, they were lysed using 2× loading buffer (100 mM Tris-HCl, 150 mM NaCl, 4% SDS, 20% glycerol). The lysates were agitated for 10 minutes at room temperature, followed by boiling at 98 °C for 10 minutes and centrifugation at 14,000 ×g for 10 minutes. The proteins were separated using 8%-12% Tris-glycine SDS-PAGE and transferred to a PVDF membrane (Millipore, 0.22 μ m). Primary antibodies were then incubated in blocking buffer overnight at 4 °C, followed by secondary antibodies at room temperature for 2 hours. The signal was detected using an Amersham Imager 600 (GE Healthcare), and densitometry was measured using ImageJ.

For immunofluorescent staining, we first washed neurons on cover glasses with PBS (supplemented with 1 mM CaCl₂ and 5 mM MgCl₂) once. Then, we permeabilized them with 0.15% Triton-X in PBS for 15 minutes. After blocked with 5% bovine serum albumin in PBS for 30 minutes at room temperature, the samples were incubated with primary antibodies in blocking buffer at 4 °C overnight. The following day, the samples were washed three times with PBS and then incubated with secondary antibodies at room temperature for 1 hour in the dark. Subsequently, the samples were washed three times with PBS and mounted in Mounting Medium with DAPI (Solarbio). Finally, images were acquired using an Andor spinning disk or Nikon confocal microscopy system. All images were captured and analyzed in a blinded manner.

Organotypic hippocampal slice culture, transfection and imaging

We prepared organotypic hippocampal slice cultures following previously established protocols (Chen et al., 2014 [\[14\]](#)). Briefly, slices were obtained from P7-8 Sprague-Dawley rats. At DIV 3, slices were biolistically transfected using a gene gun (Bio-Rad). Gold particles (1.6 μ m in diameter; Bio-Rad) were coated with 25 μ g of cDNA along with 5 μ g of GFP. Live cell imaging was conducted at DIV9/10 using a Nikon confocal system equipped with a water-immersion 60×objective. Protrusions from dendrites longer than 0.4 μ m were counted as spines. For spine density analysis, secondary basal dendrites were selected, and 2-5 different dendrites were imaged from each pyramidal cell. Both image acquisition and spine counting were carried out in a blinded manner.

A β 40 and A β 42 ELISA

We analyzed all cell sample sets using A β 40 and A β 42 ELISA kits (Invitrogen, KHB3481 and KHB3441) according to the manufacturer's instructions. The cells were sonicated in RIPA buffer (Solarbio) containing a protease inhibitor cocktail (Selleck), followed by centrifugation at 12,000 $\times$ g for 10 minutes to prepare the samples.

Lentivirus production

We inserted β -CTF-flag and GFP into the pFHTrePW vector (Ni et al., 2023 [\[1\]](#)) backbone under the control of the Tet Response Element (TRE 3G). The rtTA for the tet-off system was expressed under the control of the human synapsin I promoter. The pFHTrePW plasmid containing β -CTF-flag or GFP, along with PSPAX2 and PMD2g, was transfected into HEK293T cells. After 48 hours, the cell culture medium containing the virus was collected and filtered through a 0.45 μ m syringe filter (Millipore). Subsequently, it was concentrated using ultracentrifugation, aliquoted, and stored at -80 $^{\circ}$ C.

Adeno-Associated virus production

We performed AAV purification using triple-transfected HEK293T cells. Briefly, we transfected pHelper, pAAV2-8, and pAAV-MCS containing GFP or β -CTF into HEK293T cells at a ratio of 1:1:1. After 72 hours of incubation, the culture medium was collected in a 50 mL centrifuge tube, and the cell pellet was collected in another tube with lysis buffer (50 mM Tris-HCl, 150 mM NaCl, 2 mM MgCl₂, pH 8.0). The cell pellet was subjected to three cycles of freezing and thawing, followed by mixed with the culture medium. PEG8000 (10%) and NaCl (1 M/L) were added to the mixture. After incubating for 1 hour on ice, the mixture was centrifuged for 15 minutes at 12,000 $\times$ g and 4 $^{\circ}$ C. The precipitate was gently resuspended in lysis buffer containing DNase I (Roche) using a 1000-ml pipette. After a 30-minute incubation at 37 $^{\circ}$ C, chloroform in a 1:1 (v:v) ratio was added to each tube, shaken for 1 hour, and then centrifuged for 15 minutes at 12,000 $\times$ g and 4 $^{\circ}$ C. To harvest the concentrated AAV, the supernatant was processed using an ultrafiltration column (Millipore).

Electrophysiology

We performed patch-clamp recordings from hippocampal pyramidal cells in acute brain slices. Adult mice approximately 8 weeks old were anesthetized with isoflurane and then decapitated. The hippocampus was harvested intact and placed into cold choline cutting buffer (110 mM Choline Cl, 2.5 mM KCl, 1.25 mM NaH₂PO₄, 25 mM NaHCO₃, 7 mM MgSO₄, 25 mM D-glucose, 3.1 mM Na Pyruvate, 11.6 mM Na Ascorbate, 0.5 mM CaCl₂ at pH 7.25). Using a vibrating microtome (Leica VT1200S), the hippocampus was sliced into 400 μ m sections, followed by a 30-minute recovery period at 34 $^{\circ}$ C and another 30-minute recovery period at room temperature in artificial cerebrospinal fluid (ACSF).

The recording external solution consisted of ACSF containing (in mM): 127 NaCl, 2.5 KCl, 12.5 NaH₂PO₄, 25 NaHCO₃, 25 D-glucose, 2.5 CaCl₂, and 1.3 MgCl₂, aerated with 95% O₂/5% CO₂ to maintain a pH around 7.25. Pipettes (TW150F-4, World Precision Instruments) with a resistance of 4–6 M Ω were filled with an internal solution containing (in mM): 115 cesium methanesulfonate, 20 CsCl, 10 HEPES, 2.5 MgCl₂, 4 Na₂ATP, 0.4 Na₃GTP, 10 Na phosphocreatine, and 0.6 EGTA (pH 7.25). Miniature excitatory postsynaptic currents (mEPSCs) were recorded by whole-cell voltage clamp. The mEPSCs were recorded at a holding potential of -70 mV in the presence of 1 μ M tetrodotoxin (TTX, MedChem express, HY-12526A) and 100 μ M picrotoxin (PTX, Sigma, P1675). EPSCs were analyzed using MiniAnalysis software (Synaptosoft) with an amplitude threshold of 5 pA.

Antibodies and drugs

The following antibodies were used in this study: Y188 antibody (Abcam, ab32136), D10E5 antibody (Cell Signaling, 5606), Flag antibody (Sigma-Aldrich, F3165), β -Tubulin III antibody (Sigma, T 2200), GAPDH antibody (Proteintech, 60004-1-Ig), HA antibody (Cell Signaling, 3724S), GFP antibody (Abcam, ab13970), SYN antibody (Millipore, AB1543), GluR1 antibody (Millipore, MAB2263), GluR2 antibody (Millipore, AB1768-1). Peroxidase-Conjugated secondary antibodies: Goat Anti-Rabbit IgG (YESEN, 33101ES60), Goat Anti-Mouse IgG (YESEN, 33201ES60). Fluorescent secondary antibodies: Alexa 488 Goat anti-chicken IgG (Thermo Fisher, A11039), Alexa 568 Goat anti-chicken IgY (Thermo Fisher, A11041), Alexa 568 Goat anti-rabbit IgG (Thermo Fisher, A11036), Alexa 647 Goat anti-Mouse IgG (Thermo Fisher, A21236). Inhibitors: PF-03084014 (Selleck, S7731).

Plasmids construction

We sub-cloned the expression of GFP, BACE1-GFP, GFP-Rab5, APP, APP-IRES-BACE1 or related mutations into the pCAGGS mammalian expression vector. The AAV constructs of GFP and β -CTF-HA were sub-cloned into the pAAV-MCS vector. Lentivirus constructs of GFP, β -CTF, and β -CTF-IRES-GFP were sub-cloned into the pFHTrePW vector backbone under the control of the TRE 3G.

Behavioral Analysis

Animals underwent a 1-hour habituation period in the room before all behavioral tests. All tests were conducted and analyzed in a blinded manner.

Open field test (OFT)

The mice were introduced into a 40 × 40 cm box without roofs and permitted to explore the apparatus freely for 1 hour. Their activities were monitored using the Ethovision video tracking system (Noldus Information Technology Inc., Leesburg, VA, USA). The field was divided into a center area (24 × 24 cm) and the whole arena. Parameters such as total travel distance, duration in the center area, velocity, and rearing frequency were automatically recorded and analyzed.

Y maze

Each mouse was placed in one arm of a Y-maze (with arms measuring 30 cm in length) and allowed to explore the maze freely for 8 minutes. The movements of the mice were captured by a video camera positioned above the arena. The number of alternations and entries were analyzed using the Ethovision video tracking system.

Elevated plus maze (EPM)

Anxiety-like behavior was assessed using an elevated plus maze (EPM), which is an elevated apparatus shaped like a plus sign (+). It consists of two open arms (35 × 5 × 0.3 cm), two closed arms (36 × 5 × 18 cm) with 15 cm walls and open roofs, and a 5 × 5 cm central area. At the beginning of the test, each mouse was placed in the center of the maze facing an open arm and allowed to explore the maze freely for 5 minutes. The distance explored and time spent in the apparatus were recorded and analyzed using the SMART software (Panlab, Barcelona, Spain).

Water T maze (WTM)

Spatial learning and memory were evaluated using the Water T Maze (WTM) behavioral paradigm. In this task, mice were trained to utilize spatial cues within a room to locate a concealed platform and escape from water. In the reversal test, the hidden platform was relocated to the opposite arm to assess cognitive flexibility. The testing apparatus consisted of a plus maze, with each arm measuring 45 cm in length and 10 cm in width, constructed from clear Plexiglas. Each arm was designated as N, S, E, or W. A divider was positioned to block off an arm, allowing the

mice to choose only the E or W arm for escape. The water temperature was maintained at 25–26 °C and made opaque by adding white, nontoxic powder. An escape platform was submerged 1 cm below the water's surface on the E side of the maze, rendering it invisible to the mice.

At the beginning of each trial, the divider was put in place to block off the appropriate arm. Mice were placed at the starting point and allowed to explore the apparatus freely to reach the platform. If the mouse reached the platform within one minute, the response was considered correct; otherwise, it is considered incorrect. The experimenter recorded whether each response was correct or incorrect, and mice were permitted to remain on the platform for 10 seconds before being removed. Each day, mice underwent 5 trials with starting points semi-randomized between the N and S positions. The criterion for acquisition was achieving 80% or more correct responses averaged across the 5 trials for 2 consecutive days. After acquisition training, reversal training commenced. The hidden platform was relocated to the opposite side (W), and the same procedure was repeated until the mice successfully learned the new platform position. Mice that achieved 80% or more correct responses across the 5 trials for 2 consecutive days were deemed to have passed the test.

Fear conditioning (FC)

The fear conditioning test comprised two components: contextual fear conditioning and cued fear conditioning. During the training stage, mice were introduced into conditioning boxes and allowed to freely explore the environment for 3 minutes. The baseline freezing behavior was recorded using a visual camera and analyzed with the Ethovision video tracking system. Subsequently, mice were exposed to an 80 dB, 2 kHz tone for 30 seconds (conditioned stimulus, CS), during which an inescapable 0.3 mA foot shock (unconditioned stimulus, US) was delivered in the last 2 seconds of the tone. This procedure was repeated three times with a 30-second inter-stimulus-interval.

For the contextual fear conditioning test, conducted on the second day, mice were placed in the same conditioning box used on the previous day for 5 minutes, and their freezing behavior was recorded for further analysis. In the cued fear conditioning test, conducted on the third day after training and contextual fear conditioning, mice were placed in a different chamber from the previous one. After a 3-minute free exploration period, mice were exposed to a 3-minute tone stimulus (2 kHz, 80 dB). Freezing behavior during the 3-minute tone stimulus was measured and analyzed using the Ethovision video tracking system.

Tail suspension test (TST)

Each mouse was suspended by its tail using tape affixed to the ceiling of a three-walled rectangular compartment measuring 30 cm in height, 15 cm in width, and 15 cm in depth. The mice dangled downward, with ample space provided for movement. Video recording was conducted from the side, capturing the animals' immobile time (defined as cessation of limb movements for more than 2 seconds) during the final 4 minutes of a 6-minute session.

Stereotactic virus injection

Mice aged 4 weeks or 7–8 weeks received injections of lentivirus or AAV in the hippocampus at coordinates ML:±1.5 mm, AP: -2 mm, DV: -1.5 mm. For spine density analysis, mice were harvested after 2 months; for behavioral tests, mice were evaluated after 4 months.

Liquid chromatography tandem mass spectrometry (LC-MS/MS) sample preparation and analysis

For protein sample preparation, neurons seeded in 24-well plates underwent lysis using ultrasonic waves in RIPA buffer. After centrifugation at 18,000 ×g for 10 minutes at 4 °C, the supernatants were collected and quantified using a BCA assay (Thermo Fisher). Protein extracts were subjected

to an in-solution digest protocol, involving reduction with 5 mM Tris(2-carboxyethyl) phosphine hydrochloride (TCEP) (Aldrich, USA) followed by alkylation with 10 mM iodoacetamide (IAA) (Sigma, USA). Trypsin was added at a 1:100 ratio, and the mixture was incubated at 37 °C overnight in the dark. Digested peptides were then collected by centrifugation and desalted using C18 tips (Pierce, USA) for subsequent analysis.

For LC-MS/MS analysis, the peptide mixture was analyzed using an online EASY-nLC 1000 HPLC system coupled with an Orbitrap Fusion mass spectrometer (Thermo Fisher Scientific). The sample was loaded directly onto a 15-cm homemade capillary column (100 µm I.D., C18-AQ 1.9 µm resin, Dr. Maisch). Mobile phase A consisted of 0.1% formic acid (FA), 2% acetonitrile (ACN), and 98% H₂O, while mobile phase B comprised 0.1% FA, 2% H₂O, and 98% ACN. A 180-minute gradient (mobile phase B: 2% at 0 min, 5% at 7 min, 20% at 127 min, 35% at 167 min, 95% at 173 min, 95% at 180 min) was employed at a static flow rate of 300 nl/min.

We acquired data for proteomic analysis in a data-dependent mode, starting with one full MS1 scan in the Orbitrap (m/z: 300-1800; resolution: 120,000; AGC target value: 500,000; maximal injection time: 50 ms), followed by an MS2 scan in the linear ion trap (32% normalized collision energy; maximal injection time: 250 ms). The isolation window was set at 1.6 m/z.

Statistics

Data were processed using Microsoft Excel, and statistical analysis was performed with GraphPad Prism 7. Figures were made using Adobe Illustrator V26.3.1. Sample sizes and the statistical analyses performed are described in the respective figure legends. For all analyses, $P < 0.05$ was considered statistically significant.

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Additional information

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Conceptualization: Yelin Chen, Yang Geng.; Methodology: Yelin Chen, Yang Geng., Mengxun Luo; Investigation and Validation: Mengxun Luo., Jia Zhou., Cailu Sun, Chaoying Fu, Wanxia Chen, Chenfang Si; Visualization: Mengxun Luo; Formal analysis: Mengxun Luo, Jia Zhou, Cailu Sun, Chenfang Si, Data curation: Mengxun Luo, Chenfang Si, Yaoyang Zhang; Writing – original draft preparation: Mengxun Luo; Writing – review & editing: Yelin Chen, Yang Geng; Supervision and Project administration: Yelin Chen; Funding acquisition: Yelin Chen.

Additional files

Supplementary figures [↗](#)

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Reviewer #1 (Public review):

Summary of what the authors were trying to achieve:

In this manuscript, the authors investigated the role of β -CTF on synaptic function and memory. They report that β -CTF can trigger the loss of synapses in neurons that were transiently transfected in cultured hippocampal slices and that this synapse loss occurs

independently of A β . They confirmed previous research (Kim et al, Molecular Psychiatry, 2016) that β -CTF-induced cellular toxicity occurs through a mechanism involving a hexapeptide domain (YENPTY) in β -CTF that induces endosomal dysfunction. Although the current study also explores the role of β -CTF in synaptic and memory function in the brain using mice chronically expressing β -CTF, the studies are inconclusive because potential effects of A β generated by γ -secretase cleavage of β -CTF were not considered. Based on their findings, the authors suggest developing therapies to treat Alzheimer's disease by targeting β -CTF. While they acknowledge that clinical trials of potent BACE1 inhibitors - which also target β -CTF - have failed to show clinical improvement, their study lacks in vivo evidence directly linking β -CTF to brain function, which weakens its significance.

Major strengths and weaknesses of the methods and results:

The conclusions of the in vitro experiments using cultured hippocampal slices were well supported by the data, but aspects of the in vivo experiments need additional clarification. In contrast to the in vitro experiments in which a γ -secretase inhibitor was used to exclude possible effects of A β , this possibility was not examined in in vivo experiments assessing synapse loss and function (Fig. 3) and cognitive function (Fig. 4). The absence of plaque formation (Fig. 4C) is not sufficient to exclude the possibility that A β is involved. The potential involvement of A β is an important consideration given the 4-month duration of protein expression in the in vivo studies. This issue could be addressed using γ -secretase modulators to avoid the off-target effects of inhibitors. Evidence that the detrimental effects in mice are directly caused by β -CTF rather than indirectly via A β is critical to support the authors' conclusion.

Appraisal of whether the authors achieved their aims, and whether the results support their conclusion:

See above

Discussion of likely impact of the work on the field, and the utility of the methods and data to the community:

The authors' use of sparse expression to examine the role of β -CTF on spine loss could be a useful general tool for examining synapses in brain tissue.

Any additional context that might help readers interpret or understand the significance of the work:

The discovery of BACE1 stimulated an international effort to develop BACE1 inhibitors to treat Alzheimer's disease. BACE1 inhibitors block the formation of β -CTF which, in turn, prevents the formation of A β and other fragments. Unfortunately, BACE1 inhibitors not only did not improve cognition in patients with Alzheimer's disease, they appeared to worsen it, suggesting that β -CTF could facilitate learning and memory. Therefore, it seems unlikely that the disruptive effects of β -CTF on endosomes plays a significant role in the human disease.

Comments on revisions:

The authors may be interested in the study by Ma et al., PNAS 2007 titled "Involvement of β -site APP cleaving enzyme 1 (BACE1) in amyloid precursor protein-mediated enhancement of memory and activity-dependent synaptic plasticity," which provides significant insights into the physiological role of BACE1 in synaptic function. The researchers demonstrated that BACE1-mediated cleavage of amyloid precursor protein (APP) is essential for enhancing learning, memory, and synaptic plasticity in vivo. They observed that overexpression of APP in transgenic mice led to improved spatial memory retention and potentiation of synaptic plasticity, effects that were abolished when one or both copies of the BACE1 gene were eliminated. This suggests that BACE1's cleavage of APP facilitates activity-dependent synaptic

modifications, potentially through the production of APP intracellular domain (AICD) via β -CTF, rather than amyloid- β (A β) or soluble APP α (sAPP α). These findings highlight a physiological mechanism where BACE1-mediated APP processing leading to β -CTF supports cognitive functions, potentially explaining the detrimental effects of BACE1 inhibitors on cognitive function in clinical trials.

<https://doi.org/10.7554/eLife.100968.2.sa2>

Reviewer #3 (Public review):

Summary:

Most previous studies have focused on the contributions of Abeta and amyloid plaques in the neuronal degeneration associated with Alzheimer's disease, especially in the context of impaired synaptic transmission and plasticity which underlies the impaired cognitive functions, a hallmark in AD. But processes independent of Abeta and plaques are much less explored, and to some extent, the contributions of these processes are less well understood. Luo et al addressed this important question with an array of approaches, and their findings generally support the contribution of beta-CTF-dependent but non-Abeta dependent process to the impaired synaptic properties in the neurons. Interestingly, the above process appears to operate in a cell-autonomous manner. This cell-autonomous effect of beta-CTF as reported here may facilitate our understanding of some potential important cellular processes related to neurodegeneration. Although these findings are valuable, it is key to understand the probability of this process occurring in a more natural condition, such as when this process occurring in many neurons at the same time. This will put the authors' findings into a context for a better understanding of their contribution to either physiological or pathological processes, such as Alzheimer's. The experiments and results using cell system are quite solid, but the in vivo results are incomplete and hence less convincing (see below). The mechanistic analysis is interesting but primitive, and does not add much more weight to the significance. Hence, further efforts from the authors are required to clarify, and solidify their results, in order to provide a complete picture and support for the authors' conclusions.

Strengths:

- (1) The authors have addressed an interesting and potentially important question
- (2) The analysis using the cell system are solid and provides strong support for the authors' major conclusions. This analysis has used various technical approaches to support the authors' conclusions from different aspects and most of these results are consistent with each other.

Weaknesses:

- (1) The relevance of the authors' major findings to the pathology, especially the Abeta-dependent processes is less clear, and hence the importance of these findings may be limited.
- (2) In vivo analysis is incomplete, with certain caveats in the experimental procedures and some of the results need to be further explored to confirm the findings.
- (3) The mechanistic analysis is rather primitive and does not add further significance.

Comments on revisions:

The authors have satisfactorily addressed my main questions.

<https://doi.org/10.7554/eLife.100968.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary of what the authors were trying to achieve:

In this manuscript, the authors investigated the role of β -CTF on synaptic function and memory. They report that β -CTF can trigger the loss of synapses in neurons that were transiently transfected in cultured hippocampal slices and that this synapse loss occurs independently of A β . They confirmed previous research (Kim et al, Molecular Psychiatry, 2016) that β -CTF-induced cellular toxicity occurs through a mechanism involving a hexapeptide domain (YENPTY) in β -CTF that induces endosomal dysfunction. Although the current study also explores the role of β -CTF in synaptic and memory function in the brain using mice chronically expressing β -CTF, the studies are inconclusive because potential effects of A β generated by γ -secretase cleavage of β -CTF were not considered. Based on their findings, the authors suggest developing therapies to treat Alzheimer's disease by targeting β -CTF, but did not address the lack of clinical improvement in trials of several different BACE1 inhibitors, which target β -CTF by preventing its formation.

We would like to thank the reviewer for his/her suggestions. We have addressed the specific comments in following sections.

Major strengths and weaknesses of the methods and results:

The conclusions of the in vitro experiments using cultured hippocampal slices were well supported by the data, but aspects of the in vivo experiments and proteomic studies need additional clarification.

(1) In contrast to the in vitro experiments in which a γ -secretase inhibitor was used to exclude possible effects of A β , this possibility was not examined in in-vivo experiments assessing synapse loss and function (Figure 3) and cognitive function (Figure 4). The absence of plaque formation (Figure 4B) is not sufficient to exclude the possibility that A β is involved. The potential involvement of A β is an important consideration given the 4-month duration of protein expression in the in vivo studies.

We appreciate the reviewer for raising this question. While our current data did not exclude the potential involvement of A β -induced toxicity in the synaptic and cognitive dysfunction observed in mice overexpressing β -CTF, addressing this directly remains challenging. Treatment with γ -secretase inhibitors could potentially shed light on this issue. However, treatments with γ -secretase inhibitors are known to lead to brain dysfunction by itself likely due to its blockade of the γ -cleavage of other essential molecules, such as Notch[1, 2]. Therefore, this approach is unlikely to provide a clear answer, which prevents us from pursuing it further experimentally *in vivo*. We hope the reviewer understands this limitation. We have included additional discussion (page 14 of the revised manuscript) to highlight this question.

(2) The possibility that the results of the proteomic studies conducted in primary cultured hippocampal neurons depend in part on A β was also not taken into consideration.

We thank the reviewer for raising this question. In the revised manuscript, we examined the protein levels of synaptic proteins after treatment with γ -secretase inhibitors and found that

the levels of certain synaptic proteins were further reduced in neurons expressing β -CTF (Supplementary figure 5A-B). These results do not support A β as a major contributor of the proteomic changes induced by β -CTF.

Likely impact of the work on the field, and the utility of the methods and data to the community:

The authors' use of sparse expression to examine the role of β -CTF on spine loss could be a useful general tool for examining synapses in brain tissue.

We thank the reviewer for these comments.

Additional context that might help readers interpret or understand the significance of the work:

The discovery of BACE1 stimulated an international effort to develop BACE1 inhibitors to treat Alzheimer's disease. BACE1 inhibitors block the formation of β -CTF which, in turn, prevents the formation of A β and other fragments. Unfortunately, BACE1 inhibitors not only did not improve cognition in patients with Alzheimer's disease, they appeared to worsen it, suggesting that producing β -CTF actually facilitates learning and memory. Therefore, it seems unlikely that the disruptive effects of β -CTF on endosomes plays a significant role in human disease. Insights from the authors that shed further light on this issue would be welcome.

Response: We would like to express our gratitude to the reviewer for raising this question. It remains puzzling why BACE1 inhibition has failed to yield benefits in AD patients, while amyloid clearance via A β antibodies are able to slow down disease progression. One possible explanation is that pharmacological inhibition of BACE1 may not be as effective as its genetic removal. Indeed, genetic depletion of BACE1 leads to the clearance of existing amyloid plaques[3], whereas its pharmacological inhibition prevents the formation of new plaques but does not deplete the existing ones[4]. We think the negative results of BACE1 inhibitors in clinical trials may not be sufficient to rule out the potential contribution of β -CTF to AD pathogenesis. Given that cognitive function continues to deteriorate rapidly in plaque-free patients after 1.5 years of treatment with A β antibodies in phase three clinical studies[5], it is important to consider the potential role of other A β -related fragments in AD pathogenesis, such as β -CTF. We included further discussion in the revised manuscript (page 15 of the revised manuscript) to discuss this question.

Reviewer #2 (Public Review):

Summary:

In this study, the authors investigate the potential role of other cleavage products of amyloid precursor protein (APP) in neurodegeneration. They combine in vitro and in vivo experiments, revealing that β -CTF, a product cleaved by BACE1, promotes synaptic loss independently of A β . Furthermore, they suggest that β -CTF may interact with Rab5, leading to endosomal dysfunction and contributing to the loss of synaptic proteins.

We would like to thank the reviewer for his/her suggestions. We have addressed the specific comments in following sections.

Weaknesses:

Most experiments were conducted in vitro using overexpressed β -CTF. Additionally, the study does not elucidate the mechanisms by which β -CTF disrupts endosomal function and induces synaptic degeneration.

We would like to thank the reviewer for this comment. While a significant portion of our experiments were conducted *in vitro*, the main findings were also confirmed *in vivo* (Figure 3 and 4). Repeating all the experiments *in vivo* would be challenging and may not be possible because of technical difficulties. Regarding the use of overexpressed β -CTF, we acknowledge that this represents a common limitation in neurodegenerative disease studies. These diseases progress slowly over decades in patients. To model this progression in cell or mouse models within a time frame feasible for research, overexpression of certain proteins is often inevitable. Since β -CTF levels are elevated in AD patients[6], its overexpression is not a irrelevant approach to investigate its potential effects.

We did not further investigate the mechanisms by which β -CTF disrupted endosomal function because our preliminary results align with previous findings that could explain its mechanism. Kim et al. demonstrated that β -CTF recruits APPL1 (a Rab5 effector) via the YENPTY motif to Rab5 endosomes, where it stabilizes active GTP-Rab5, leading to pathologically accelerated endocytosis, endosome swelling and selectively impaired transport of Rab5 endosomes[6]. However, this paper did not show whether this Rab5 overactivation-induced endosomal dysfunction leads to any damages in synapses. In our study, we observed that co-expression of Rab5_{S34N} with β -CTF effectively mitigated β -CTF-induced spine loss in hippocampal slice cultures (Figures 6L-M), indicating that Rab5 overactivation-induced endosomal dysfunction contributed to β -CTF-induced spine loss. We included further discussion in the revised manuscript to clarify this (page 15 of the revised manuscript).

Reviewer #3 (Public Review):

Summary:

Most previous studies have focused on the contributions of Abeta and amyloid plaques in the neuronal degeneration associated with Alzheimer's disease, especially in the context of impaired synaptic transmission and plasticity which underlies the impaired cognitive functions, a hallmark in AD. But processes independent of Abeta and plaques are much less explored, and to some extent, the contributions of these processes are less well understood. Luo et al addressed this important question with an array of approaches, and their findings generally support the contribution of beta-CTF-dependent but non-Abeta-dependent process to the impaired synaptic properties in the neurons. Interestingly, the above process appears to operate in a cell-autonomous manner. This cell-autonomous effect of beta-CTF as reported here may facilitate our understanding of some potentially important cellular processes related to neurodegeneration. Although these findings are valuable, it is key to understand the probability of this process occurring in a more natural condition, such as when this process occurs in many neurons at the same time. This will put the authors' findings into a context for a better understanding of their contribution to either physiological or pathological processes, such as Alzheimer's. The experiments and results using the cell system are quite solid, but the in vivo results are incomplete and hence less convincing (see below). The mechanistic analysis is interesting but primitive and does not add much more weight to the significance. Hence, further efforts from the authors are required to clarify and solidify their results, in order to provide a complete picture and support for the authors' conclusions.

We would like to thank the reviewer for the suggestions. We have addressed the specific comments in following sections.

Strengths:

(1) The authors have addressed an interesting and potentially important question

(2) The analysis using the cell system is solid and provides strong support for the authors' major conclusions. This analysis has used various technical approaches to support the authors' conclusions from different aspects and most of these results are consistent with each other.

We would like to thank the reviewer for these comments.

Weaknesses:

(1) The relevance of the authors' major findings to the pathology, especially the Abeta-dependent processes is less clear, and hence the importance of these findings may be limited.

We would like to thank the reviewer for this question. Phase 3 clinical trial data from A β antibodies show that cognitive function continues to decline rapidly, even in plaque-free patients, after 1.5 years of treatment[5]. This suggests that plaque-independent mechanisms may drive AD progression. Therefore, it is crucial to consider the potential contributions of other A β species or related fragments, such as alternative forms of A β and β -CTF. While it is early to predict how much β -CTF contributes to AD progression, it is notable that β -CTF induced synaptic deficits in mice, which recapitulates a key pathological feature of AD. Ultimately, the contribution of β -CTF in AD pathogenesis can only be tested through clinical studies in the future.

(2) In vivo analysis is incomplete, with certain caveats in the experimental procedures and some of the results need to be further explored to confirm the findings.

We would like to thank the reviewer for this suggestion. We have corrected these caveats in the revised manuscript.

(3) The mechanistic analysis is rather primitive and does not add further significance.

We would like to thank the reviewer for this comment. We did not delve further into the underlying mechanisms because our analysis indicates that Rab5 overactivation-induced endosomal dysfunction underlies β -CTF-induced synaptic dysfunction, which is consistent with another study and has been addressed in our study[6]. We hope the reviewer could understand that our focus in this paper is on how β -CTF triggers synaptic deficits, which is why we did not investigate the mechanisms of β -CTF-induced endosomal dysfunction further.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Suggestions for improved or additional experiments, data, or analyses:

(1) In Figures 4H, 4J, 4K and Supplemental Figures 3C, 3E, and 3G, it was unclear whether a repeated measures 2-way ANOVA, rather than a 2-way ANOVA, followed by appropriate post-hoc analyses was used to strengthen the conclusion that there were significant effects in the behavioral tests.

We appreciate the reviewer for raising this point and apologize for the lack of clear description in the manuscript. In those figures mentioned above, we use a repeated measures 2-way ANOVA to analyze the data by Graphpad Prism. In Figure 4H, fear conditioning tests were conducted. The same cohort of mice were used in the baseline, contextual and cued tests. Firstly, baseline freezing was tested; then these mice underwent tone and foot shock

training, followed by contextual test and cued test. So, a repeated measures 2-way ANOVA is more appropriate for the experiment.

In water T maze tests (Figure 4J and K), the same cohort of mice were trained and tested each day. So, it's also appropriate to use a repeated measures 2-way ANOVA.

In Supplementary figure 3C, 3E and 3G, OFT was conducted. In this experiment, the locomotion of the same cohort of mice were recorded. Also, it's appropriate to use a repeated measures 2-way ANOVA.

Clearer description for these experiments has been provided in the revised manuscript.

(2) Including gender analyses would be helpful.

The mice we used in this study were all males.

Minor corrections to text and figures:

(1) Quantitative analyses in Figures 5A-C, 5H, 6G, 6H, and Supplementary Figures 4 and 5C would be helpful.

We have provided quantitative analysis of these results (Figure 5D, 5J, 6K, Supplementary figure 4D, 5F) mentioned above in the revised manuscript.

(2) Percent correct (%) in Figures 4J and 4K should be labeled as 0, 50, and 100 instead of 0.0, 0.5, and 1.0.

We would like to thank the reviewer for pointing out this. We have made corrections in the revised manuscript.

Reviewer #2 (Recommendations For The Authors):

In the study conducted by Luo et al, it was observed that the fragment of amyloid precursor protein (APP) cleaved by beta-site amyloid precursor protein cleaving enzyme 1 (BACE1), known as β -CTF, plays a crucial role in synaptic damage. The study found increasing expression of β -CTF in neurons could induce synapse loss both in vitro and in vivo, independent of $A\beta$. Mechanistically, they explored how β -CTF could interfere with the endosome system by interacting with RAB5. While this study is intriguing, there are several points that warrant further investigation:

(1) The study involved overexpressing β -CTF in neurons. It would be valuable to know if the levels of β -CTF are similarly increased in Alzheimer's disease (AD) patients or AD mouse models.

We would like to thank the reviewer for the suggestion. It's reported β -CTF levels were significantly elevated in the AD cerebral cortex[6]. Most AD mouse models are human APP transgenic mouse models with elevated β -CTF levels[7].

(2) The study noted that β -CTF in neurons is a membranal fragment, but the overexpressed β -CTF was not located in the membrane. It is important to ascertain whether the membranal β -CTF and cytoplasmic β -CTF lead to synapse loss in a similar manner.

We apologize for not clearly explaining the localization of β -CTF in the original manuscript. β -CTF is produced from APP through β -cleavage, a process that occurs in organelles such as

endo-lysosomes[8]. The overexpressed β -CTF is also primarily localized in the endo-lysosomal systems (Figure 5C and Supplementary figure 4C), similar to those generated by APP cleavage.

(3) The study found a significant decrease in GluA1, a subunit of AMPA receptors, due to β -CTF. It would be beneficial to investigate whether there are systematic alterations in NMDA receptors, including GluN2A and GluN2B.

We would like to express our gratitude to the reviewer for bringing up this question. The protein levels of GluN2A and GluN2B are also reduced in neurons expressing β -CTF (Figure 6E-F)

(4) The study showed a significant decrease in the frequency of miniature excitatory postsynaptic currents (mEPSC), indicating disrupted presynaptic vesicle neurotransmitter release. It would be pertinent to test whether the expression level of the presynaptic SNARE complex, which is required for vesicle release, is altered by β -CTF.

We would like to express our gratitude to the reviewer for bringing up this question. The protein level of the presynaptic SNARE complex, such as VAMP2, is also reduced in neurons expressing β -CTF (Figure 6E, G).

(5) Since AMPA receptors are glutamate receptors, it is important to determine whether the ability of glutamate release is altered by β -CTF. In vivo studies using a glutamate sensor should be conducted to examine glutamate release.

We would like to express our gratitude to the reviewer for this suggestion. It will be interesting to use glutamate sensors to assess the ability of glutamate release in the future.

(6) The quality of immunostaining associated with Figures 4B and 4C was noted to be suboptimal.

We apologize for the suboptimal quality of these images. The immunostaining in Figures 4B and 4C were captured using the stitching function of a confocal microscope to display larger areas, including the entire hemisphere and hippocampus. We have reprocessed the images to obtain higher-quality versions.

(7) It would be insightful to investigate whether treatment with a BACE1 inhibitor in the study could reverse synaptic deficits mediated by β -CTF.

We would like to thank the reviewer for this suggestion. In Figure 11I-M, we constructed an APP mutant (APP_{MV}), which cannot be cleaved by BACE1 to produce β -CTF and A β but has no impact on β' -cleavage. When co-expressed with BACE1, APP_{MV} failed to induce spine loss, supporting the effect of β -CTF. We think these results demonstrate that β -CTF underlies the synaptic deficits. It would be interesting to test the effects of BACE1 inhibition in the future.

(8) Considering the potential implications for therapeutics, it is worth exploring whether extremely low levels of β -CTF have beneficial effects in regulating synaptic function or promoting synaptogenesis at a physiological level.

We would like to thank the reviewer for raising this question. We found that when the plasmid amount was reduced to 1/8 of the original dose, β -CTF no longer induced a decrease in dendritic spine density (Supplementary figure 2E-F). It's reported APP-Swedish mutation in familial AD increased synapse numbers and synaptic transmission, whereas inhibition of BACE1 lowered synapse numbers, suppressed synaptic transmission in wild type neurons, suggesting that at physiological level, β -CTF might be synaptogenic[9].

(9) *The molecular mechanism through which β -CTF interferes with Rab5 function should be elucidated.*

We would like to thank the reviewer for raising this question. Kim et al have elucidated the mechanism through which β -CTF interferes with Rab5 function. β -CTF recruited APPL1 (a Rab5 effector) via YENPTY motif to Rab5 endosomes, where it stabilizes active GTP-Rab5, leading to pathologically accelerated endocytosis, endosome swelling and selectively impaired transport of Rab5 endosomes[6]. We have included additional discussion for this question in the revised manuscript (page 15 of the revised manuscript).

(10) *The study could compare the role of β -CTF and $A\beta$ in neurodegeneration in AD mouse models.*

We would like to thank the reviewer for raising this point. While it is easier to dissect the role of $A\beta$ and β -CTF *in vitro*, some of the critical tools are not applicable *in vivo*, such as γ -secretase inhibitors, which lead to severe side effects because of their inhibition on other γ substrates[1, 2]. Therefore it will be difficult to demonstrate their different roles *in vivo*. There are studies showing that β -CTF accumulation precedes $A\beta$ deposition in model mice and mediates $A\beta$ independent intracellular pathologies[10, 11], consistent with our results.

(11) *Based on the findings, it would be valuable to discuss possible explanations for the failure of most BACE1 inhibitors in recent clinical trials for humans.*

Response: We would like to express our gratitude to the reviewer for raising this recommendation. It is a big puzzle why BACE1 inhibition failed to provide beneficial effects in AD patients whereas clearance of amyloid by $A\beta$ antibodies could slow down the AD progress. One potential answer is that pharmacological inhibition of BACE1 might be not as effective as its genetic removal. Indeed, genetic depletion of BACE1 leads to clearance of existing amyloid plaques[3], whereas pharmacological inhibition of BACE1 could not stop growth of existing plaques, although it prevents formation of new plaques[4]. The negative result of BACE1 inhibitors might not be sufficient to exclude the possibility that β -CTF could also contribute to the AD pathogenesis. We have included additional discussion for this question in the revised manuscript (page 15 of the revised manuscript).

Reviewer #3 (Recommendations For The Authors):

Major:

(1) *The cell experiments were performed at DIV 9, do the authors know whether at this age, the neurons are still developing and spine density has not reached a plateau yet? If so, the observed effect may reflect the impact on development and/or maturation, rather than on the mature neurons. The authors should be more specific about this issue.*

We would like to thank the reviewer for pointing out this question. These slice cultures were made from 1-week-old rats. DIV 9 is about two weeks old. These neurons are still developing and spine density has not reached a plateau yet[12]. In addition, we also investigated the effects of β -CTF on the synapses of mature neurons in two-month-old mice (Figure 3). So we think the observed effect reflects the impact on both immature and mature neurons.

(2) *mEPSCs shown in Figure 3D were of small amplitudes, perhaps also indicating that these synapses are not yet mature.*

In Figure 3D, the mEPSC results were obtained from pyramidal neurons in the CA1 region of two-month-old mice. At the age of two months, neurotransmitter levels and synaptic density

have reached adult levels[13].

(3) There was no data on the spine density or mEPSCs in the mice OE b-CTF, hence it is unclear whether a primary impact of this manipulation (b-CTF effect) on the synaptic transmission still occurs in vivo.

In Figure 3, we examined the density of dendritic spines and mEPSCs from CA1 pyramidal neurons infected with lentivirus expressing β -CTF in mice and showed that those neurons expressing additional amount of β -CTF exhibited lower spine density and less mEPSCs, supporting that β -CTF also damaged synaptic transmission *in vivo*.

(4) OE of b-CTF should lead to the production of Abeta, although this may not lead to the formation of significant plaques. How do the authors know whether their findings on behavioral and cognitive impairments were not largely mediated by Abeta, which has been widely reported by previous studies?

We would like to thank the reviewer for pointing out this question. Indeed, our *in vivo* data could not exclude the potential involvement of A β in the pathology, despite the absence of amyloid plaque formation. It will be difficult to demonstrate this question *in vivo* because of the severe side effects from γ inhibition.

(5) Figure 4H, the freezing level in the cued fear conditioning was very high, likely saturated; this may mask a potential reduction in the b-CTF OE mice (there is a hint for that in the results). The authors should repeat the experiments using less strong footshock strength (hence resulting in less freezing, <70%).

We would like to express our gratitude to the reviewer for bringing up this question. The contextual fear conditioning test assesses hippocampal function, while the cued fear conditioning test assesses amygdala function. We hope the reviewer understands that our primary goal is to assess hippocampus-related functions in this experiment and we did see a significant difference between GFP and β -CTF groups. Therefore, we think the intensity of footshock we used was suitable to serve the primary purpose of this experiment.

(6) Why was the deficit in the Morris water maze in the b-CTF OE mice only significant in the training phase?

We would like to thank the reviewer for raising this question and apologize for not describing the test clearly. This is a water T maze test, not Morris water maze test.

To make the behavioral paradigm of the water T maze test easier to understand, we have provided a more detailed description of the methods in the new version of the manuscript.

The acquisition phase of the Water T Maze (WTM) evaluates spatial learning and memory, where mice use spatial cues in the environment to navigate to a hidden platform and escape from water, while the reversal learning measures cognitive flexibility in which mice must learn a new location of the hidden platform[14]. In reversal learning task (Figure 4J-K), the learning curves of the two groups of mice did not show any significant differences, indicating that the expression of β -CTF only damages spatial learning and memory but not cognitive flexibility. This is consistent with a previous report using APP/PS1 mice[15].

(7) Will the altered Rab5 in the b-CTF OE condition also affect the level of other proteins?

We would like to express our gratitude to the reviewer for raising this interesting question. Expression of Rab5_{S34N} in β -CTF-expressing neurons did not alter the levels of synapse-

related proteins that were reduced in these neurons (Supplementary figure 5G-H), suggesting Rab5 overactivation did not contribute to these protein expression changes induced by β -CTF.

(8) How do the authors reconcile their findings with the well-established findings that Abeta affects synaptic transmission and spine density? Do they think these two processes may occur simultaneously in the neurons, or, one process may dominate in the other?

APP, A β , and presenilins have been extensively studied in mouse models, providing convincing evidence that high A β concentrations are toxic to synapses[16]. Moreover, addition of A β to murine cultured neurons or brain slices is toxic to synapses[17]. However, A β -induced synaptotoxicity was not observed in our study. A major difference between our study and others is that our study used a isolated expression system that apply A β only to individual neurons surrounded by neurons without excessive amount of A β , whereas the rest studies generally apply A β to all the neurons. Therefore, we predict that A β does not lead to synaptic deficits from individual neurons in cell autonomous manners, whereas β -CTF does. A β and β -CTF represent two parallel pathways of action. Additional discussion for this question has been included in the revised manuscript (page 14 of the revised manuscript).

Minor:

Fig 2F-G, "prevent" rather than "reverse"?

We would like to thank the reviewer for pointing this out. We have made corrections in the revised manuscript.

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